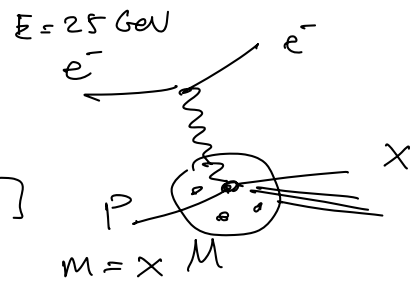


Deep Inelastic Scattering (DIS) e-p



$$\left. \frac{d\sigma}{dQ^2} \right|_{\text{DIS}} = \frac{\alpha^2}{Q^4} E'^2 \frac{E'}{E} \left[\underbrace{W_2(Q^2, \nu)}_{\cos^2 \frac{\theta}{2}} + \underbrace{2W_1(Q^2, \nu)}_{\sin^2 \frac{\theta}{2}} \right]$$

e- proton elastic scattering

Spin = 1/2

$x \in [0, 1]$

$$\left. \frac{d\sigma}{dQ^2 dE'} \right|_{eq} = \frac{\alpha^2}{Q^4} E'^2 \frac{E'}{E} Z_q^2 \left[\checkmark \cos^2 \frac{\theta}{2} + 2 \frac{Q^2}{4m^2} \sin^2 \frac{\theta}{2} \right] \delta\left(\nu - \frac{Q^2}{2m}\right)$$

condition for elastic scattering.
assumption in my calculation

$$W_1(Q^2, \nu) \Big|_x = Z_q^2 \frac{Q^2}{4m^2} \delta\left(\nu - \frac{Q^2}{2m}\right)$$

for $x = \nu/M$

Introduce parton density function $\int_0^1 f(x) dx = 1$

$$W_1(Q^2, \nu) = Z_q^2 \int_0^1 dx f(x) \frac{Q^2}{4m^2} \delta\left(\nu - \frac{Q^2}{2m}\right)$$

$$\delta\left(\nu - \frac{Q^2}{2m}\right) \Rightarrow \nu = \frac{Q^2}{2m} = \frac{Q^2}{2xM} \Rightarrow x_0 = \frac{Q^2}{2M\nu}$$

$$\delta\left(\nu - \frac{Q^2}{2Mx}\right) = \delta(g(x)) \quad x \in (0, 1)$$

$$\begin{aligned} W_1(Q^2, \nu) &= Z_q^2 \int_0^1 dx f(x) \frac{Q^2}{4x^2 M^2} \delta(g(x)) \\ &= Z_q^2 \frac{Q^2}{4M^2} \int_0^1 dx f(x) \frac{1}{x^2} \delta(g(x)) \end{aligned}$$

$$\int A(x) \delta(g(x)) dx = A(x_0) \frac{1}{|g'(x)|_{x_0}}$$

$$g(x) = \nu - \frac{Q^2}{2Mx}$$

$$g'(x) = \frac{Q^2}{2M} \frac{1}{x^2}$$

$$x_0 = \frac{Q^2}{2M\nu}$$

$$g'(x_0) = \frac{Q^2}{2M} \frac{4M^2 \nu^2}{Q^4} = \frac{2M\nu^2}{Q^2}$$

$$W_1(Q^2, \nu) = Z_q^2 \frac{Q^2}{4M^2} f(x_0) \frac{1}{x_0^2} \frac{1}{|g'(x_0)|} \quad x = \frac{Q^2}{2M\nu}$$

$$= Z_q^2 \frac{\cancel{Q^2}}{\cancel{4M^2}} \frac{\cancel{4M^2\nu^2}}{\cancel{Q^2}} \frac{\cancel{Q^2}}{2M\nu^2} f(x)$$

$$= Z_q^2 \frac{f(x)}{2M} \Rightarrow W_1(Q^2, \nu) = W_1\left(\frac{Q^2}{2M\nu}\right)$$

$$W_2(Q^2, \nu) = \int_0^1 dx f(x) Z_q^2 \delta\left(\nu - \frac{Q^2}{2Mx}\right) = Z_q^2 \frac{x}{\nu} f(x)$$

Structure functions:

$$W_1(Q^2, \nu) = \frac{Z_q^2}{2M} f(x) \quad \text{For 1 parton.}$$

$$W_2(Q^2, \nu) = \frac{Z_q^2}{\nu} x f(x).$$

For $n > 1$ partons.

$$W_1(Q^2, \nu) = \frac{1}{2M} \sum_i^{\text{partons}} Z_{q_i}^2 f_i(x)$$

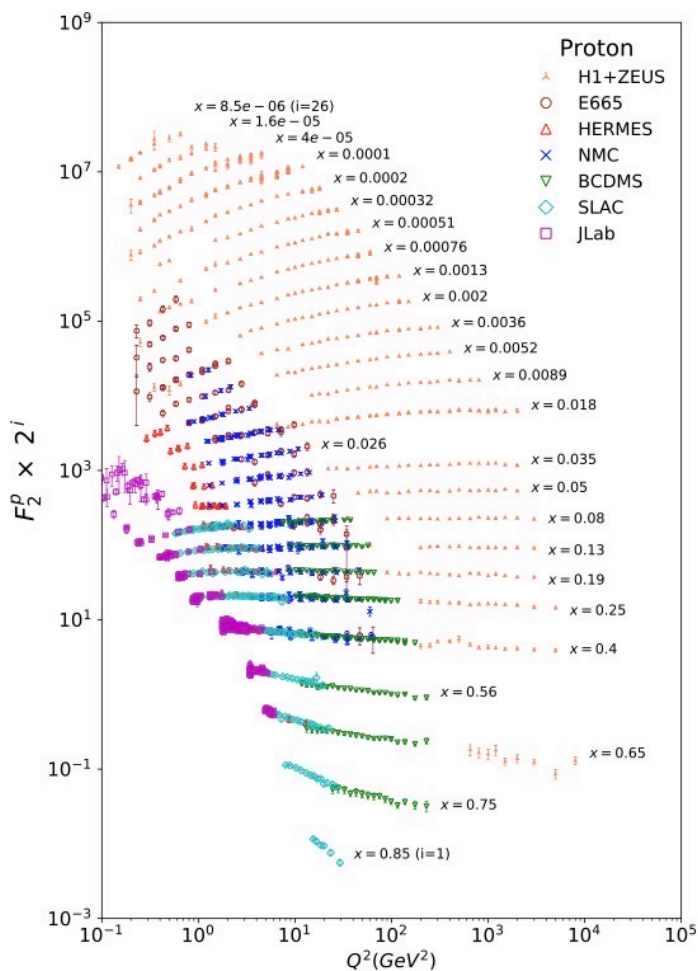
$$W_2(Q^2, \nu) = \frac{1}{\nu} \sum_i^{\text{partons}} Z_{q_i}^2 x f_i(x)$$

$$F_1(x) = M W_1(Q^2, \nu) = \frac{1}{2} \sum_i Z_{q_i}^2 f_i(x)$$

$$F_2(x) = \nu W_2(Q^2, \nu) = \sum_i Z_{q_i}^2 x f_i(x)$$

$$\Rightarrow \boxed{F_2(x) = 2x F_1(x)}$$

$f_i(x)$: parton density functions for $n > 1$ partons.



$$Q^2 > 4EE' \sin^2 \frac{\theta}{2}$$

$$\nu = E - E'$$

you measure $(E', \theta) \rightarrow (Q^2, \nu)$.

$$x = \frac{Q^2}{2M\nu}$$

$\Rightarrow F_1(x)$ do not depend on Q^2

$\Rightarrow w_1(Q^2, \nu)$ does not depend on Q^2, ν separately.

$$F_2(x) = 2x F_1(x)$$

Callan-Gross equation

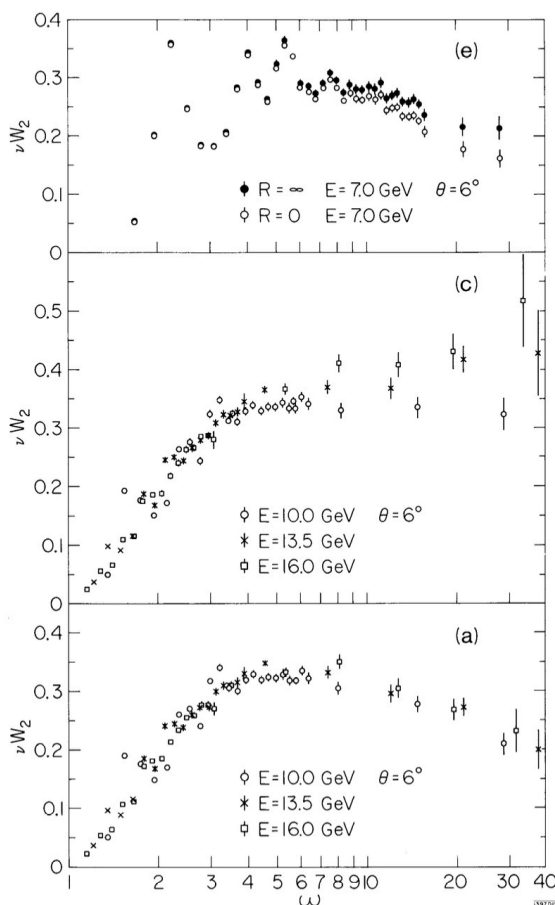


Fig. 2a, 2c, 2e

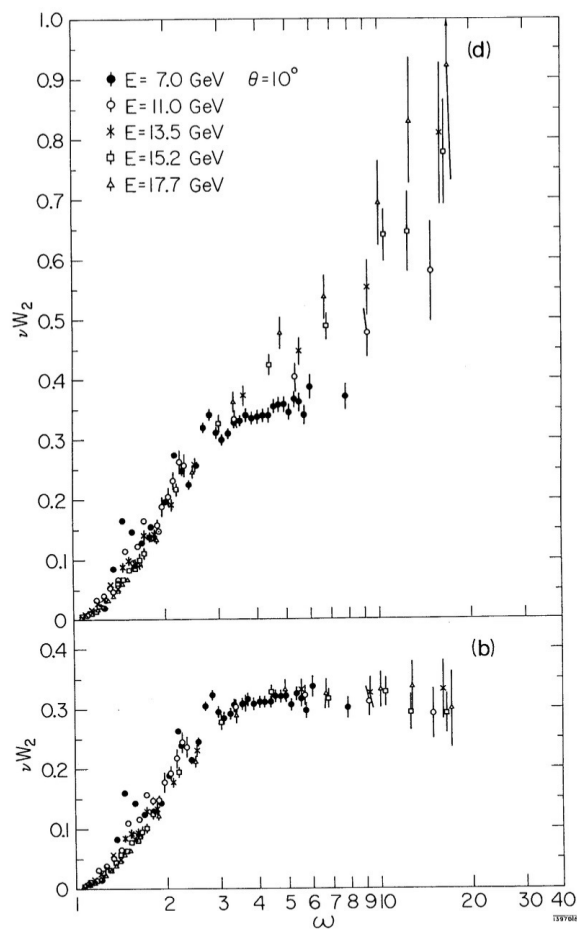
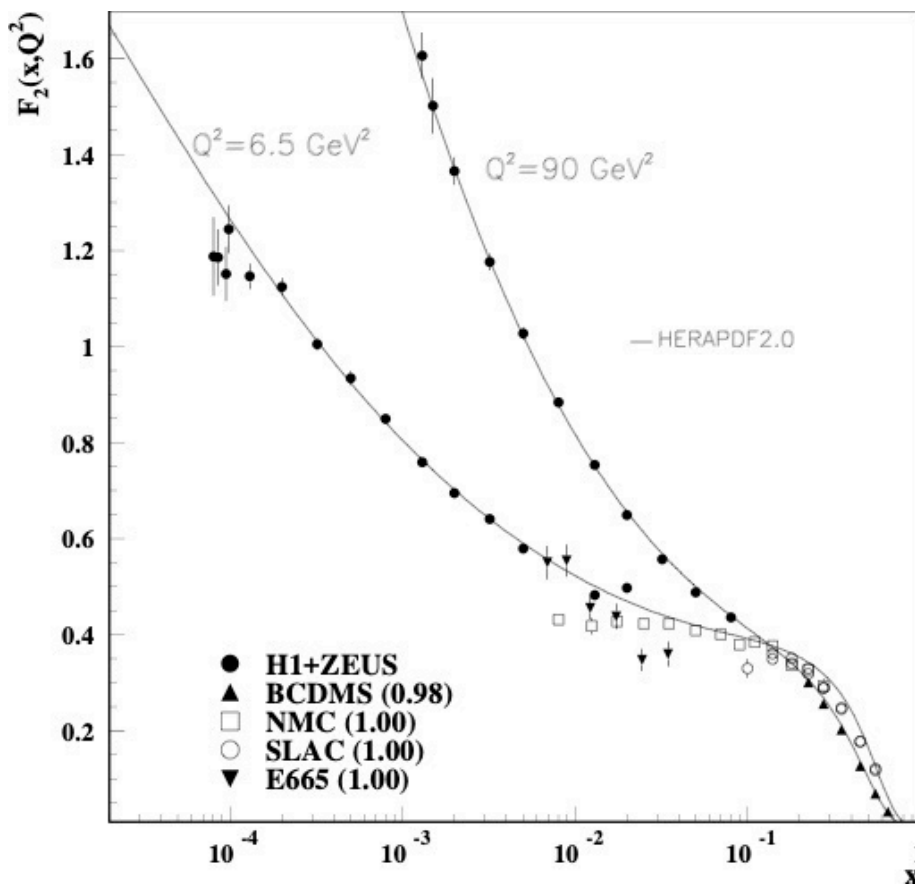


Fig. 2b, 2d

$$\omega = \frac{1}{x}$$

$$= \frac{2M\nu}{Q^2}$$

$$\nu W_2 = F_2$$

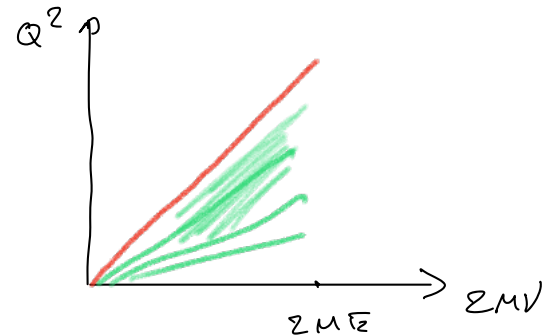


$$x_B = \frac{Q^2}{2m\nu}$$

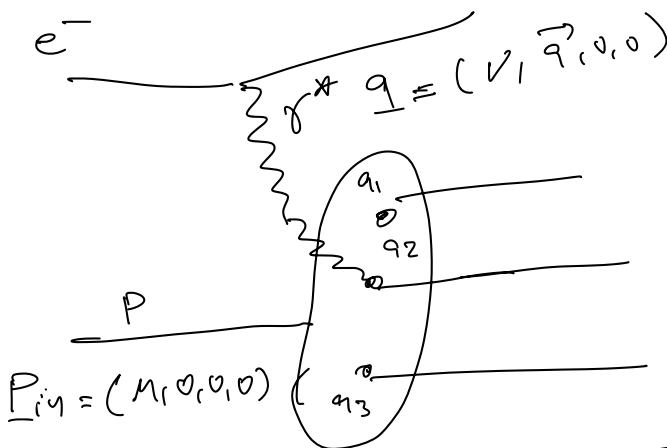
Computed only from e^-

Bjorken x

$$x_B = \frac{4EE' \sin^2 \frac{\theta}{2}}{2M(E-E')}$$



Feynman x



In DIS we assume γ^* interacts with a single parton/quark.

$$\longleftrightarrow x$$

$$\underline{P}_q^{fin} = \underline{P}_q^{in} + \underline{q}$$

$$E_e = 25 \text{ GeV}$$

$$\underline{P}_{in} = \sum_i \underline{P}_{q_i}$$

$$\underline{P}_q = x_F \underline{P}_{in}$$

Fraction of momentum carried by the quark/parton.

In high energy regime $|\underline{P}_L| \gg |\underline{P}_T|$

$$|\underline{P}_q^{fin}|^2 = (\underline{P}_q^{in} + \underline{q})^2 = (\underline{P}_q^{in})^2 + (\underline{q})^2 + 2 \underline{P}_q^{in} \cdot \underline{q}$$

$$m_q^2 = m_q^2 - Q^2 + 2 \underline{P}_q^{in} \cdot \underline{q}$$

In high energy regime assume $m_q \approx 0$ negligible mass.

$$Q^2 = 2 \frac{P_{in}^2}{s} \cdot q = 2 x_F \frac{P_{in}^2}{s} \cdot q = 2 x_F M \nu$$

$$x_F = \frac{Q^2}{2M\nu} \quad \text{fraction of momentum carried by the parton.}$$

In high energy limit $x_F \equiv x_B$

$\Rightarrow$ partons exist with $z_{q_i} < 1$

how many? what z_q ? $\Rightarrow e^+e^-$ collisions

$$F_1(x) = M W_1(Q^2, \nu) = \frac{1}{2} \sum_i z_{q_i}^2 f_i(x)$$

$$F_2(x) = \nu W_2(Q^2, \nu) = \sum_i z_{q_i}^2 x f_i(x)$$

$$F_2(x) = 2x F_1(x)$$

$f_i(x)$: pdf (parton density function)

For quark of type i

z_{q_i} : charge of quark:

$\sum_i$: sum over types of partons

$$\begin{pmatrix} u \\ d \end{pmatrix} \begin{pmatrix} c \\ s \end{pmatrix} \begin{pmatrix} t \\ b \end{pmatrix} \begin{matrix} +2/3 \\ -1/3 \end{matrix}$$

$$p = (uud) \quad Q_p = +2 = \frac{2}{3} + \frac{2}{3} - \frac{1}{3} = +1$$

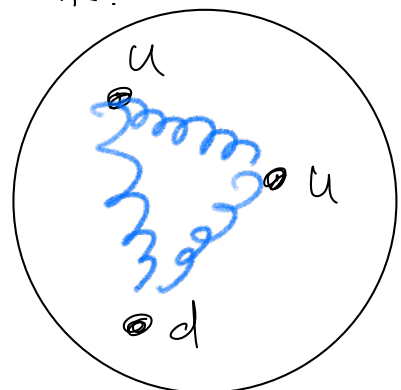
Experimentally measure $\sum_i x_i = ?$

$$P_{\text{prot}} = \sum_i P_{q_i} = \sum_i x_i P_{q_i}$$

Conservation of 4-mom $\Rightarrow \sum_i x_i = 1$.

Experimentally $\sum_i x_i = 0.5$

uud : valence quarks.



$$p = \underbrace{u + \bar{u} + d + \bar{d}}_{\text{valence quarks}} + \underbrace{\bar{u} + \bar{d} + d + \bar{d} + s + \bar{s} + g_1 + g_2}_{\text{sea quarks and gluons}}$$

$\sum_{\text{sea}} q_i = 0$

→ provide electric charge.

Sea: made of $q_i \bar{q}_i$ pairs + gluons

HERA Collider @ DESY

$\xrightarrow{p}$
 $E = 920 \text{ GeV}$

$\xleftarrow{e^-}$

27.5 GeV

$$P_p = (920, 920, 0, 0) \quad P_e = (27.5, -27.5, 0, 0)$$

$$P_{\text{tot}} = (947.5, 892.5, 0, 0)$$

$$\sqrt{s} = \sqrt{(P_{\text{tot}})^2} = 320 \text{ GeV} \quad \text{access high } Q^2$$

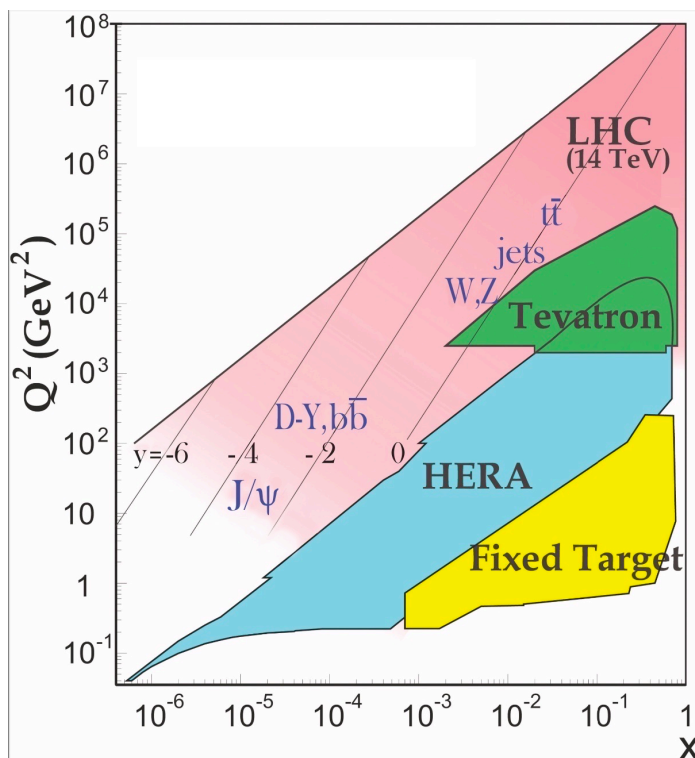
SLAC

$\xrightarrow{\quad}$
 $P_e (25, 25, 0, 0)$

$\xrightarrow{\quad}$

$P_p = (11, 0, 0, 0)$

$$\sqrt{s} = \sqrt{(25)^2 - (25)^2}$$



$p + p$ @ $\sqrt{s} = 13 \text{ TeV}$

Tevatron $p + \bar{p}$ $\sqrt{s} = 1.96 \text{ TeV}$

HERA $\sqrt{s} = 320 \text{ GeV}$

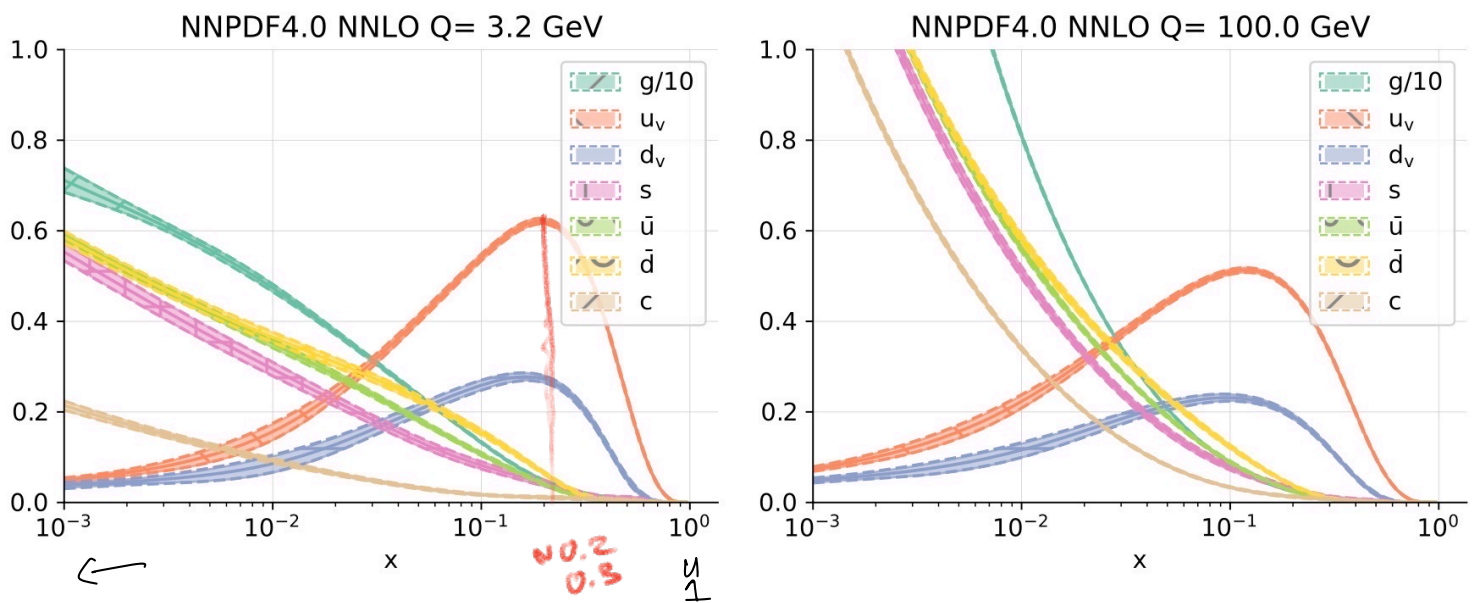


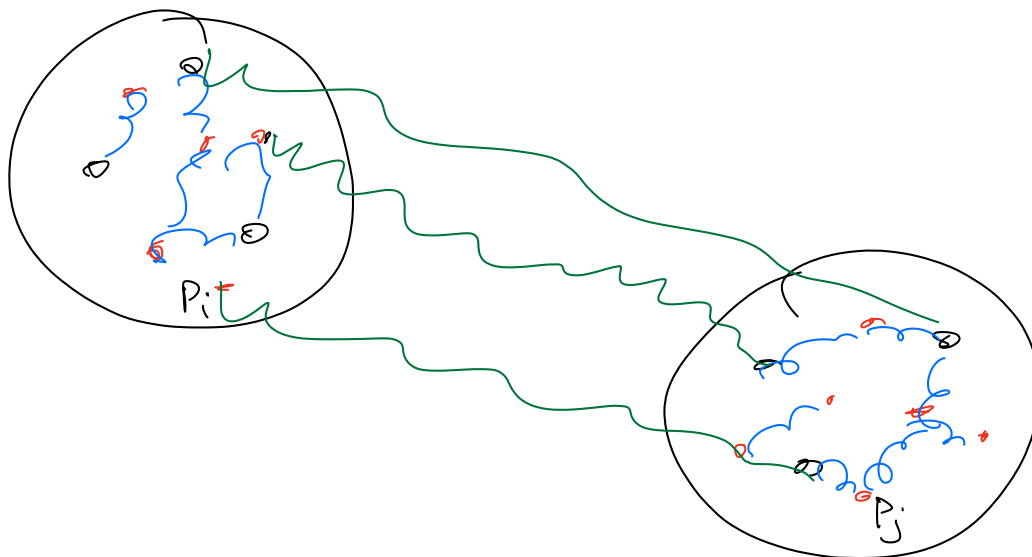
Figure 1.1. The NNPDF4.0 NNLO PDFs at $Q = 3.2$ GeV (left) and $Q = 10^2$ GeV (right).

valence quark carries $\sim \frac{1}{3} P_{\text{prot}}$

For $x \rightarrow 0$ gluon parton density function dominant

u_v : u quark p.d.f. valence quark.

$\bar{u}$: p.d.f. for $\bar{u}$ from the sea.



$$\sigma(p+p \rightarrow X)$$

$$\sum_i^{\text{proton 1}} \sum_j^{\text{proton 2}} \int dx_i dx_j f_i(x_i) f_j(x_j) \sigma(P_i + P_j \rightarrow X)$$

Σ over all
partons
at all x_i, x_j

hard scattering between parton like
partons (valence or sea partons)

neutron p.d.f $\Rightarrow$

proton p.d.f

$n = (udd)$

$p = (uud)$

Different valence quarks.

$u_V^n(x)$

$u_V^p(x)$

$d_V^n(x)$

$d_V^p(x)$

$u_{Sea}^n(x)$

$u_{Sea}^p(x)$

Sum Rules or Constraints over P.d.f.

$f(x)$: parton density function

$$x = \frac{|P_{q1}|}{|P_{had}|}$$

$$\sum_i P_{q_i} = P_{had}$$

$$= \sum_i x_i P_{had} = P_{had} = \sum_i x_i = 1$$

$$\sum_i \int_0^1 dx f_i(x) x = 1$$

Mom. Sum rule

$$\int_0^1 dx \left[\sum_{\text{flavors}} (q_f(x) + \bar{q}_f(x)) + \sum_{\text{gluons}} g(x) \right] x = 1$$

Valence quark $\Rightarrow$ provides electric charge.

proton $\int_0^1 dx [u(x) - \bar{u}(x)] = 2$

$$Q_p = 1$$

$$\int_0^1 dx [d(x) - \bar{d}(x)] = 1$$

$$= \underbrace{\frac{2}{3} + \frac{2}{3}}_{u=2} - \underbrace{\frac{1}{3}}_{d=1}$$